

Python: warm-up exercises on branching

This notebook was generated in solutions mode.

Exercise: Check if Number is Even

Exercise Description

Write a piece of code that checks whether a number `num` is even. If so, print a message.

Your solution

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise: Even or Odd Message

Exercise Description

Write a piece of code that checks whether a number `num` is even. Print a message about the outcome (odd or even).

Your solution

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
else:
    print("Condition evaluates to False.")
    print("Number", num, "is odd")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise: Conditional assignment

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable `result`, depending on the outcome of the condition. Print the outcome at the end.

Your solution

```
num = 42
if num % 2 == 0:
    result = "even"
else:
    result = "odd"
print("Number", num, "is", result)
```

Debug your solution

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```
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```

Exercise: Multiple Divisibility Check

Exercise Description

Write a piece of code that checks whether a number `num` is divisible by 2, 3, 5 and 7. Print a message when `num` is divisible by one of this number.

Your solution

```
num = 42
if num % 2 == 0:
    print("Number ", num, "is divisible by 2.")
if num % 3 == 0:
    print("Number ", num, "is divisible by 3.")
if num % 5 == 0:
    print("Number ", num, "is divisible by 5.")
```

```
if num % 7 == 0:  
    print("Number ", num, "is divisible by 7.")
```

Debug your solution

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```
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```

Exercise: Nested Conditional Logic

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

Your solution

```
num = 15  
if num % 2 == 0:  
    print("Number", num, "is even.")  
    if num % 3 == 0:
```

```
        print("Number", num, "is also divisible by 3.")
    else:
        print("Number", num, "is not divisible by 3.")
else:
    print("Number", num, "is odd.")
    if num % 5 == 0:
        print("Number", num, "is also divisible by 5.")
    else:
        print("Number", num, "is not divisible by 5.")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise: Dog Age Calculator

Exercise Description

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

Your solution

```
age = int(input("Enter the dog's age in human years: "))
print("Inserted age:", age)

if age <= 2:
    dog_years = age * 10.5
else:
    dog_years = 2 * 10.5 + (age - 2) * 4

print("The dog's age in dog's years is:", dog_years)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

